

KRUSHANG PRAJAPATI

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Surat, Gujarat | DOB: 30 May 2006

PROFILE OVERVIEW

Second-year Electronics and Communication Engineering student (Sem 4) with hands-on experience in embedded systems, IoT development, and digital electronics. Skilled in working with ESP32, Firebase-based dashboards, circuit simulation using LTSpice, and Linux environments.

Interested in hardware–software integration, AI-based vision systems, FPGA/VLSI design, and cybersecurity projects. Passionate about building practical systems that combine electronics, cloud platforms, and intelligent automation.

EDUCATION

B.Tech — Electronics and Communication Engineering (Pursuing)

Sarvajanik College of Engineering & Technology, Sarvajanik University, Surat

2024 – 2028

SPI: 7.3/10 (Sem 4)

HSC — Class XII, Science (PCM)

BhagwanMahavir School, Surat | CBSE

2024

64%

SSC — Class X

B.S Patel Primary School, Bilimora | CBSE

2022

79%

SKILLS

Embedded & IoT: ESP32, Arduino, ESP32-CAM, Sensor Integration, Relay Modules, IoT System Design, IoT Dashboards

Electronics & Hardware: Analog & Digital Electronics, Op-Amps, Diodes, Transistors, Circuit Design & Prototyping

Circuit Simulation: LTSpice, MATLAB Simulink, Vivado

Programming: C, Python, MATLAB

Cloud & Backend: Firebase Realtime Database, Firebase Authentication, Backend Integration

Systems: Linux (Arch Linux & Ubuntu), Git, GitHub, Command Line Tools

Machine Learning: Machine Learning Fundamentals, Motion Detection, Computer Vision

Soft Skills: Critical Thinking, Leadership, Communication, Team Collaboration

ACADEMIC PROJECTS

FPGA-Based Image Processing System

Academic Project | Verilog HDL | Xilinx Vivado | Basys3 FPGA (Artix-7) | [GitHub](#)

- Designed and implemented an image processing system on the Basys3 (Artix-7) FPGA in Verilog HDL, applying 8 image filters in hardware.
- Won 1st place at the FPGA Design Challenge 2.0, SCET, Surat.

Direct-Mapped Write-Back Cache Controller

Academic Project | Verilog HDL | Xilinx Vivado | Basys3 FPGA | [GitHub](#)

- Designed a direct-mapped, write-back, write-allocate cache controller in Verilog HDL with a 5-state FSM (IDLE, COMPARE, WRITE_BACK, MEM_FETCH, UPDATE), 256 cache lines and 4-word line size.
- Verified with a testbench; added a Basys3 top-level (switches/LEDs) to demonstrate hits, misses and write-backs.

FPGA-Based Digital Design Implementations

Academic Project | Verilog HDL | Xilinx Vivado | Basys3 FPGA

- Implemented Sequence Detector (Moore & Mealy FSM), Sequence Generator, arithmetic units (Adder / Subtractor), Flip-Flops, MUX, and DSD modules in Verilog HDL — validated on FPGA.

ESP32-CAM AI Motion Detection & Monitoring System

Academic Project | ESP 32 - CAM | FireBase | AI Motion Detection

- Developed a smart surveillance system using ESP32-CAM integrated with an AI-based motion detection model. Detected movement through camera input and sent alerts/data to a Firebase realtime dashboard for remote monitoring.

XOR-Based Symmetric Encryption Engine

Academic Project | Verilog HDL | FPGA

- Implemented parameterised encryption / decryption hardware with configurable key width and key-locked decryption logic — validated on FPGA.

POSITIONS OF RESPONSIBILITY

Google Student Ambassador

Aug 2025 – Jan 2026

Hybrid Mode

- Explored Gemini AI capabilities; conducted sessions training college students on using AI tools for academic and professional workflows.

IEEE Student Member | T&P Cell Volunteer

2024 – 2025

ACHIEVEMENTS

- 1st Prize FPGA Design Challenge, SCET, Surat
- 3rd Prize Networking using Cisco Packet Tracer, SCET, Surat
- Won Chess Tournament, SCET, Surat
- Participated International Press, SUMUN 2025, Sarvajanik University
- Volunteer — Cyber Workshop, Sarvajanik University, Surat
- Volunteer — SAW 3.0 2025, Pre-Symposium, SCET